

Question-4 (10 marks)

Choose the correct answers:

- 1 Which of the following is a solution?
(A) Iron (B) Water (C) Alloy of Brass (D) O₂ gas
- 2 Which one of the following concentration units varies with temperature?
(A) Molality (B) Mole fraction (C) Molarity (D) All of the above
- 3 A solution that contains the maximum amount of a solute in a given solvent, at a specific temperature:
(A) Concentrated (B) Unsaturated (C) Dilute (D) Saturated
- 4 The scientific principle that the volume of a fixed quantity of gas at constant temperature is inversely proportional to the pressure is called:
(A) Avogadro's Law (B) Gay-Lussac's Law (C) Boyle's Law (D) Charles's Law
- 5 Heat is measured in...
(A) Joules (B) Calories (C) Kilocalories (D) Any of these
- 6 What is the correct formula to convert Celsius to Kelvin?
(A) $K = (C - 273)$ (B) $K = (C + 273)$ (C) $K = (C - 100)$ (D) $K = (C + 100)$
- 7 The lead storage battery, in such cells the:
(A) Electrical energy is stored as Chemical energy (B) Chemical energy is stored as heat energy
(C) Kinetic energy is stored as potential energy (D) All of the above
- 8 Liquid fuels are mainly:
(A) Coal and wood (B) Natural gas (C) Oils and petroleum products (D) wood only
- 9 When a solute dissolves in water, it is expected to
(A) Raise the freezing point of water (B) Lower the boiling point of the water
(C) Raise the boiling point of the water (D) All of the above.
- 10 The decrease in the freezing point of a solution depends on:
(A) The number of solute particles in the solution (B) Pressure (C) Volume (D) Work
- 11 The petrochemical products are
(A) Plastics (B) rubbers (C) fertilizers (D) A, B and C
- 12 The amount of energy required to raise the temperature of one gram of pure water by one degree Celsius is defined as:
(A) The specific heat (B) Heat capacity (C) Temperature (D) Pressure
- 13 Molality is defined as the:
(A) Moles solute/moles solvent (B) Moles solute/Liters solution
(C) Moles solute/kg solvent (D) Moles solute/kg solution
- 14 $(\Delta U = q + W)$ in this formula W stands for:
(A) Change in Internal Energy (B) Heat capacity (C) Work (D) Pressure
- 15 In Van der Waals equation, the constant a and b?
(A) Are equal to each other for any real gas (B) Are equal to 1 for ideal gases
(C) Are used to correct for the difference between Celsius and Kelvin
(D) Are used to correct for the volume of gas molecules and the attractive forces between gas molecules.

3/5

- | | | |
|--|--------------------|--|
|  BUC
BAHRAIN UNIVERSITY IN CARBON
جامعة البحرين بالسياسة | 2024-2025 | ana Technology |
| | Final Written Exam | |
| | Date: 13/1/2025 | |
| | | Course Code: BAS 041
Course Level: 0
Exam Hours: 3 |
- 16 The specific heat of water is 4.18 J/g°C. The specific heat of Iron is 0.45 J/g°C. What material needs more heat energy to raise the temperature?
(A) Water (B) Iron (C) Both are the same (D) I cannot determine
 - 17 Under what set of conditions a gas would be expected to behave as real gas?
(A) High temperature and low pressure. (B) High temperature and high pressure.
(C) Low temperature and high pressure. (D) Low temperature and low pressure.
 - 18 Which of the following gases will have the highest diffusion rate with temperature held constant? (molecular weight: CO₂=44, NH₃=17, SO₂=64, O₂=32)
(A) O₂ (B) NH₃ (C) CO₂ (D) SO₂
 - 19 Sand is insoluble in water. The solubility of sand in 100 g of water at 20 °C is...
(A) 0 g (B) 20 g (C) 100 g (D) 120
 - 20 If a balloon of Argon is heated what will happen to the volume?
(A) Volume will increase (B) Volume will decrease
(C) Volume will not change (D) I cannot determine

100%

A. (2 marks)
A 466-g sample of water is heated from 8.50°C to 74.60°C. Calculate the amount of heat absorbed (in kilojoules) by the water. Specific heat of water is 4.184 J/g°C.

B. (2 marks)

Calculate the ratio of the effusion of Nitrogen gas to the ratio of effusion of Oxygen. (molar mass: N=14, O=16).

$$\frac{R_1}{R_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{32}{28}} = 1.06$$

C. (2 marks)

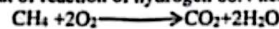
What is the osmotic pressure (in atmospheres) of a 12.36 M aqueous urea solution at 22.0°C.
R = 0.082057 L.atm/(mol.K)

D. (2 marks)

Draw and write the equation represent the reactions that occur in galvanic cell.

E. (2 marks)

Calculate heat of combustion of methane CH₄ if heat of formation 18.4 kcal. (Heat of reaction of carbon = 94.4 kcal and heat of reaction of hydrogen 68.4 kcal.)



Question-4 (10 marks)

Choose the correct answers:

1. Heat is measured in...

- (a) Joules (b) Calories (c) Kilocalories (d) Any of these

2. What is the correct formula to convert Celsius to Kelvin?

- (a) K = (C-273) (b) K = (C+273) (c) K = (C-100) (d) K = (C+100)

2/4

 BUC BADR UNIVERSITY IN CAIRO جامعة بدر بالقاهرة	Fall Semester 2022-2023 Final Written Exam Date: 16/1/2023	Faculty of Engineering and Technology Course Code: BAS 041 Course Level: 0 Exam Hours: 3
--	--	--

3. Galvanic cell is a device that changes:

- (a) Chemical energy into electrical energy (b) Kinetic energy to potential energy
(c) Chemical energy to heat energy (d) All of the above

4. Liquid fuels are mainly:

- (a) Coal. (b) Natural gas (c) Oils and petroleum products (d) wood

5. When a solute dissolves in water, it is expected to

- (a) Raise the freezing point of water (b) Lower the boiling point of the water
(c) Lower the freezing point of water (d) All of the above.

6. Specific heat of water is 4.18 J/g°C. Specific heat of wood is 1.760 J/g°C. What material needs more heat energy to raise the temperature?

- (a) Water (b) Wood (c) Both are same (d) I cannot determine

7. Under what set of conditions a gas would be expected to behave as real gas?

- (a) High temperature and low pressure. (b) High temperature and high pressure.
(c) Low temperature and high pressure. (d) Low temperature and low pressure.

8. If a balloon of helium is heated what will happen to the volume?

- (a) Volume will increase (b) Volume will decrease
(c) Volume will not change (d) I cannot determine

9. The petrochemicals products are

- a) Plastics b) rubbers c) fertilizers d) a,b and c

10. The amount of energy required to raise the temperature of one gram of pure water by one degree Celsius is defined as:

- (a) The specific heat (b) Heat capacity (c) Temperature (d) Pressure

11. Molality is defined as the:

- (a) Moles solute/moles solvent (b) Moles solute/Liters solution
(c) Moles solute/kg solvent (d) Moles solute/kg solution

12. ($\Delta U = q + W$) in this formula W stands for

- (a) Change in Internal Energy (b) Heat capacity (c) Work (d) Pressure

13. In Van der Waals equation, the constant a and b?

- (a) Are equal to each other for any real gas
(b) Are equal to 1 for ideal gases
(c) Are used for correcting for the difference between Celsius and Kelvin
(d) Are used to correct for the volume of gas molecules and the attractive forces between gas molecules.

14. Which of the following gases will have the lowest diffusion rate with temperature held constant? (Molecular weight: CO₂ = 44, NH₃ = 17, SO₂ = 64, O₂ = 32)

- (a) O₂ (b) NH₃ (c) CO₂ (d) SO₂

3/4

- (a) High temperature and low pressure. (b) High temperature and high pressure.
(c) Low temperature and high pressure. (d) Low temperature and low pressure.
8. If a balloon of helium is heated what will happen to the volume?
(a) Volume will increase (b) Volume will decrease
(c) Volume will not change (d) I cannot determine
9. The petrochemicals products are
a) Plastics b) rubbers c) fertilizers d) a, b and c
10. The amount of energy required to raise the temperature of one gram of pure water by one degree Celsius is defined as:
(a) The specific heat (b) Heat capacity (c) Temperature (d) Pressure
11. Molality is defined as the:
(a) Moles solute/moles solvent (b) Moles solute/Liters solution
(c) Moles solute/kg solvent (d) Moles solute/kg solution
12. $\Delta U = q + W$ in this formula W stands for
(a) Change in Internal Energy (b) Heat capacity (c) Work (d) Pressure
13. In Van der Waals equation, the constant a and b?
(a) Are equal to each other for any real gas
(b) Are equal to 1 for ideal gases
(c) Are used for correcting for the difference between Celsius and Kelvin
(d) Are used to correct for the volume of gas molecules and the attractive forces between gas molecules.
14. Which of the following gases will have the lowest diffusion rate with temperature held constant?
(Molecular weight: $\text{CO}_2 = 44$, $\text{NH}_3 = 17$, $\text{SO}_2 = 64$, $\text{O}_2 = 32$)
(a) O_2 (b) NH_3 (c) CO_2 (d) SO_2

3/4

 BUC BACH UNIVERSITY IN CAIRO جامعة باح	Fall Semester 2022-2023 Final Written Exam Date: 16/1/2023	Faculty of Engineering and Technology Course Code: BAS 041 Course Level: 0 Exam Hours: 3
---	--	--

15. Rubber is insoluble in water. The solubility of rubber in 100 g of water at 20 °C is...
(a) 0 g (b) 20 g (c) 100 g (d) 120 g

16. Which of the following is not a solution?
(a) Salt water (b) Air (c) Alloy of Brass (d) O_2 gas

17. Which one of the following concentration units varies with temperature?
(a) Molality (b) Mole fraction
(c) Molarity (d) All of the above

18. A solution containing the little amount of dissolved solute is:
(a) Concentrated. (b) Unsaturated
(c) Dilute. (d) Saturated.

19. Most of gases become soluble in water as the:
(a) Pressure is increased. (b) Temperature is decreased.
(c) Temperature is increased. (d) Pressure is decreased.

20. The scientific principle that, the volume of a fixed quantity of gas at constant temperature is inversely proportional to the pressure is called:
(a) Avogadro's Law (b) Gay-Lussac's Law (c) Boyle's Law (d) Charles's Law

Question-5 (10 marks)
Put True (✓) or false (X)

- Boiling point elevation is one of the colligative properties. ✓
- The freezing point depression of a solution is dependent on the number of the solute particles in solution. ✓
- Portland cement is used in lining chemical apparatuses. ✓
- A solution may exist as a gas, liquid or solid. ✓
- The boiling point of a solution is always higher than that of a pure solvent. ✓
- High alumina cement is used in the manufacture of kilns and furnaces up to 1800° C. ✓
- For most substances, increasing the temperature of the solvent increasing the solubility of the solute. ✓
- The van der Waals equation describes the properties of real gases under certain conditions. ✓
- At constant pressure, the volume of an ideal gas is inversely proportional to the temperature. ✓
- At constant temperature, the volume of an ideal gas is inversely proportional to the pressure. ✓
- Isolated systems that cannot exchange either energy or matter with the surroundings. ✓
- The first law of thermodynamics states that: energy can be converted from one form to another but cannot be created or destroyed. ✓
- The solubility of a gas in a solution increase as its pressure decreases. ✓
- Petrochemical Industries needs an integrated petrochemical company. ✓
- The solubility of a gas dissolved in a liquid increase as the temperature of the solution increases. ✓
- It is recommended to use Portland slag cement in reinforced concrete. ✓
- The rate of corrosion decreases with temperature increase. ✓
- The endothermic reaction is the reaction where the system absorbs the energy from its surrounding in the form of heat. ✓
- Metals subjected to bending stresses, hammering and strains are not corroding. ✓
- A battery is a group of galvanic cells connected in series or in parallel. ✓

With best wishes
Dr. Ashraf Sheraton

4/4